

Final 2026 IS & IT & CS

Faculty of Computers and Information

Information Technology Department

Computer Networks (3rd Year IT, IS & CS)

Academic Year: 2025-2026

Final Exam (60 Marks)

Part I: Choose the right answers (Q1–Q50 × 0.8 = 40 Marks)

1. What is the standard number for Token Ring technology defined by the IEEE?

- (A) 802.3
- (B) 802.11
- (C) 802.15
- (D) 802.5

2. What does the Type/Length field in an Ethernet frame specify when its value is less than 1536?

- (A) The protocol type
- (B) The length of the payload
- (C) The frame priority
- (D) The checksum value

3. How many bits are used to represent the host portion of an IP address in Class B?

- (A) 8 bits
- (B) 24 bits
- (C) 32 bits
- (D) 16 bits

4. In control plane, the remote controller computes and installs forwarding tables in routers.

- (A) per-router
- (B) DSL
- (C) QoS
- (D) SDN

5. In DNS records, if the type is MX, the name is, and the value is

- (A) domain, the hostname of the authoritative name server
- (B) hostname, name of SMTP mail server associated with it

- (C) hostname, IP address
- (D) alias name, canonical name

6. What is the purpose of routing metrics?

- (A) To calculate the shortest path between nodes
- (B) To determine the network topology
- (C) To prioritize certain types of traffic
- (D) To prevent routing loops

7. Which of the following fields comes first in an Ethernet frame?

- (A) Destination MAC Address
- (B) Type / Length
- (C) Source MAC Address
- (D) Frame Check Sequence (FCS)

8. Which routing protocol is an exterior gateway protocol used for routing between different Autonomous Systems (ASes)?

- (A) RIP
- (B) OSPF
- (C) BGP
- (D) EIGRP

9. UDP can be used for all of the following, except

- (A) DNS
- (B) HTTP/3
- (C) SNMP
- (D) IMAP

10. Which layer handles the formatting, encryption, and compression of data in the OSI model?

- (A) Physical layer
- (B) Presentation layer
- (C) Data link layer
- (D) Session layer

11. Which layer of the OSI model does the Ethernet frame belong to?

- (A) Data Link layer
- (B) Transport layer
- (C) Network layer
- (D) Physical layer

12. protocol is responsible for error reporting and router signaling.

- (A) IP

- (B) ICMP
- (C) OSPF
- (D) BGP

13. In control plane, individual routing algorithm components in every router interact in the control plane.

- (A) SDN
- (B) DSL
- (C) QoS
- (D) per-router

14. The 'Protocol' field in the IPv4 header indicates

- (A) The transport layer protocol used in the data portion
- (B) The type of network layer protocol
- (C) The type of data link layer protocol
- (D) The routing algorithm used

15. different channels transmitted in different frequency bands.

- (A) FDM
- (B) FDD
- (C) TDM
- (D) TDD

16. permit sites to learn a lot about the user on their site.

- (A) Sessions
- (B) Databases
- (C) Cookies
- (D) Web caches

17. All of the following are the main features of SMTP, except:

- (A) It has ASCII command/response interaction and status codes
- (B) Client pull
- (C) Multiple objects sent in a multipart message
- (D) The server uses CRLF.CRLF to determine the end of the message

18. What does the throughput rate represent in a network?

- (A) The highest and lowest frequencies in the network
- (B) The rate of incoming data at a specific point in the network
- (C) The capacity of the network's servers and workstations
- (D) The speed at which information can be sent between devices

19. In, the underlying channel can have errors and lose packets (data, ACKs).

- (A) rdt2.2

- (B) rdt2.0
- (C) rdt3.0
- (D) rdt2.1

20. How many bytes does a MAC address contain?

- (A) 4 bytes
- (B) 6 bytes
- (C) 10 bytes
- (D) 8 bytes

21. In IPv6 datagram format, all of the following are missing compared with IPv4, except

- (A) flow label
- (B) checksum
- (C) options
- (D) fragmentation/reassembly

22. is the main component of the network core that concerns the network interconnection.

- (A) server
- (B) host
- (C) hub
- (D) router

23. What does the prefix of an IP address identify?

- (A) The host within the network
- (B) The network to which the computer is connected
- (C) The geographical location of the computer
- (D) The protocol used for communication

24. Which protocol is used by the Novell Netware operating system for data transfer within networks?

- (A) ARP
- (B) RARP
- (C) IPX/SPX
- (D) ICMP

25. Which IP address class is used for multicast?

- (A) Class A
- (B) Class B
- (C) Class C
- (D) Class D

26. Consider sending over HTTP/2 a Web page that consists of one video clip, and five images. Suppose that the video clip is transported as 2000 frames, and each image has three frames. If all the video frames are sent first without interleaving, how many "frame times" are needed until all five images are sent?

- (A) 2000
- (B) 2030
- (C) 4030
- (D) 2015

27. What is the purpose of the Border Gateway Protocol (BGP)?

- (A) Routing within an Autonomous System (AS)
- (B) Routing between different Autonomous Systems (ASes)
- (C) Load balancing within a network
- (D) Prioritizing certain types of traffic

28. is a standards organization for the Internet and is responsible for the technical standards that make up the Internet protocol suite (TCP/IP).

- (A) RFC
- (B) IETF
- (C) ISP
- (D) CMTS

29. What is the default port for FTP?

- (A) 21
- (B) 25
- (C) 53
- (D) 80

30. Which field in an IPv4 header specifies the length of the entire packet, including header and data?

- (A) Header Length
- (B) Total Length
- (C) Identification
- (D) Fragment Offset

31. What is X.25?

- (A) A network protocol for local area networks
- (B) A network protocol for wide area networks
- (C) A file transfer protocol
- (D) An email communication protocol

32. What is the purpose of DNS in a computer network?

- (A) To convert domain names to IP addresses

- (B) To encrypt data transmission
- (C) To manage network security
- (D) To establish wireless connections

33. Which protocol supports packets containing error messages, control, and information?

- (A) ARP
- (B) UDP
- (C) ICMP
- (D) RARP

34. All of the following are the main features of the UDP, except

- (A) No connection establishment
- (B) No connection state at sender and receiver
- (C) Large header size
- (D) No congestion control

35. The non-persistent HTTP response time equals to

- (A) as little as one RTT for all the referenced objects
- (B) as little as two RTT for all the referenced objects
- (C) 2RTT
- (D) 2RTT+ file transmission time

36. How do nodes in a distance vector algorithm update their routing tables?

- (A) By sending broadcast messages
- (B) By exchanging routing tables periodically
- (C) By using multicast delivery
- (D) By manually configuring routes

37. All communication activity on the Internet is governed by

- (A) services
- (B) ISP
- (C) gateway
- (D) protocols

38. Which protocol is commonly used by routers to exchange routing information?

- (A) TCP
- (B) UDP
- (C) DNS
- (D) RIP

39. What is the purpose of NAT (Network Address Translation)?

- (A) To translate domain names to IP addresses
- (B) To map private IP addresses to public IP addresses

- (C) To encrypt network traffic
- (D) To assign IP addresses to devices

40. provide(s) encrypted TCP connections, data integrity, and end-point authentication.

- (A) TCP sockets
- (B) TLS
- (C) Vanilla TCP & UDP sockets
- (D) UDP sockets

41. The typical transmission rate for wide-area cellular access networks is

- (A) 1 Gbps
- (B) 100 Kbps
- (C) 10's Mbps
- (D) 10's Gbps

42. Which protocol is used to determine the logical address when the physical address is known?

- (A) ARP
- (B) RARP
- (C) ICMP
- (D) UDP

43. method includes user data in the URL field of the HTTP request message following a '?'

- (A) GET
- (B) HEAD
- (C) PUT
- (D) POST

44. Using all devices in the local network share just one IPv4 address as far as the outside world is concerned.

- (A) DHCP
- (B) WFQ
- (C) CIDR
- (D) NAT

45. The main difference that distinguishes HTTP/2 over the other earlier versions is

- (A) The server responds in order (FCFS) to GET requests.
- (B) The connection is persistent.
- (C) It adds security, per object error- and congestion control (more pipelining) over UDP.
- (D) Transmission order of requested objects based on client-specified object priority.

46. manages root DNS domain.

- (A) ICANN
- (B) DNSSEC
- (C) Comcast
- (D) Akamai

47. What is the advantage of a tree network topology?

- (A) Point-to-point wiring for individual segments
- (B) Direct connectivity between devices without relying on a central device
- (C) Easy to install and wire
- (D) Scalability for future network expansion

48. handles data from multiple sockets and adds a transport header.

- (A) Multiplexing
- (B) Demultiplexing
- (C) Routing
- (D) Forwarding

49. topology is used to interconnect the routers in the network.

- (A) Mesh
- (B) Bus
- (C) Ring
- (D) Star

50. In the router, longest prefix matching is often performed using It retrieves addresses in one clock cycle, regardless of table size.

- (A) CAM
- (B) RED
- (C) ECN
- (D) TCAMs

Part II: Answer the following questions (20 Marks)

Q2 [6 Marks]: A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, another packet is halfway through transmission on this outbound link, and five packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,200 bytes and the link rate is 2.1 Mbps. What is the packet's queuing delay?

Q3 [7 Marks]: An ISP is granted a block of IP addresses starting with 172.20.0.0/15 (131,072 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:
(a) First Group: 128 customers (Each customer needs 512 IP addresses)

(b) Second Group: 256 customers (Each customer needs 128 IP addresses)

(c) Third Group: 64 customers (Each customer needs 256 IP addresses)

Design appropriate CIDR subblocks for each group. Clearly show the address ranges allocated to each group. Calculate the number of unused IP addresses after all allocations.

Q4 [7 Marks]: Users are generating data at a rate of 200 kbps when busy, but are busy generating data only with probability ($p = 0.1$). Suppose that we have a 10 Gbps link. What is N , the maximum number of users that can be supported simultaneously under circuit switching?
